

IoT SuperServer SYS-221HE-FTNRD

X13 DP 2U Short-depth front I/O System with 6 NVMe/SATA bays and redundant DC PSU



More details here

Key Applications

Telecom Micro Data Center, AI Inference and Machine Learning, Cloud Computing, Network Function Virtualization (NFV), 5G Networking,

Key Features

- Dual Socket E (LGA-4677) 5th/4th Gen Intel® Xeon® Scalable processors;
- 32 DIMM slots supporting up to 8TB of memory; RDIMMs up to DDR5-5600;
- Configurable PCIe slot options up to 8 PCIe 5.0 x8 (6 FHFL + 2 FHHL) or 4 PCIe 5.0 x16 (3 FHFL + 1 FHHL);
- Flexible networking options with 2 AIOM networking slots (OCP NIC 3.0 compatible);
- Up to 6x 2.5" hot-swap drive bays with options to support NVMe/SATA drives;
- 6 heavy duty hot-swap fans with optimal fan speed control;
- Redundant 1300W -48V DC power supplies;



Form Factor	2U Rackmount Enclosure: 436.88 x 88.9 x 574mm (17.2" x 3.5" x 22.6") Package: 598 x 247 x 938mm (23.5" x 9.7" x 36.9")
Processor	Dual Socket E (LGA-4677) 5th Gen Intel® Xeon®/4th Gen Intel® Xeon® Scalable processors Supports Intel Xeon CPU Max Series with high bandwidth memory (HBM) Up to 64C/128T; Up to 320MB Cache per CPU
GPU	Max GPU Count: Up to 3 double-width or 3 single-width GPUs Supported GPU: NVIDIA PCIe: H100, A10, A2, A40, L40, L40S Intel PCIe: Intel Data Center GPU Flex 170 CPU-GPU Interconnect: PCIe 5.0 x16 CPU-to-GPU Interconnect GPU-GPU Interconnect: PCIe
System Memory	Slot Count: 32 DIMM slots/16 Channels Max Memory (2DPC): Up to 8TB 5600MT/s ECC DDR5 RDIMM
Drive Bays Configuration	Default: Total 6 bays • 6 front hot-swap 2.5" NVMe/SAS*/SATA drive bays (*SAS support may require additional storage controller and/or cables) M.2: 2 M.2 NVMe/SATA slots (M-key 2280/22110)
Expansion Slots	Default • 3 PCIe 5.0 x16 FHFL slots • 2 PCIe 5.0 x8 FHHL slots • 2 PCIe 5.0 x16 AIOM slots (OCP 3.0 compatible) Option A • 6 PCIe 5.0 x8 FHFL slots • 2 PCIe 5.0 x8 FHHL slots • 2 PCIe 5.0 x16 AIOM slots (OCP 3.0 compatible) Option B • 3 PCIe 5.0 x16 FHFL slots • 1 PCIe 5.0 x16 FHHL slot • 2 PCIe 5.0 x16 AIOM slots (OCP 3.0 compatible)
On-Board Devices	SATA: SATA (6Gbps) ; RAID 0/1/5/10 support NVMe: NVMe; RAID 0/1/5/10 support(Intel® VROC RAID key required) Network Connectivity: 2 SFP28 25GbE with Broadcom® BCM57414 (optional) 2 QSFP28 100GbE with Broadcom® BCM57508 (optional) 2 QSFP28 100GbE with Intel® E810-CAM2 (optional) 2 QSFP28 100GbE (optional) 4 SFP+ 10GbE with Intel® X710-BM2 (optional) 2 QSFP56 200GbE (optional)

2 QSFP56 100GbE (optional)
2 RJ45 1GbE with Intel® I350-AM2 (optional)
4 SFP+ 10GbE with Intel® X710-TM4 (optional)
1 OSFP 400GbE with NVIDIA® ConnectX®-7 NDR (optional)
4 RJ45 10GbE with Intel® X550 (optional)

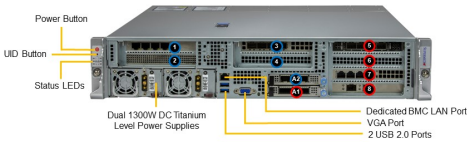
Input / Output	LAN: 1 RJ45 1 GbE Dedicated BMC LAN port
	USB: 2 USB 2.0 ports(front)
	Video: 1 VGA port

(Rear View – System)



Drive Bay	Description
1 – 4	4 Hot-swap 2.5" NVMe/SATA3 Drive Bays (NVMe from CPU1)
5 – 6	2 Hot-swap 2.5" NVMe/SATA3 Drive Bays (NVMe from CPU2)

(Front View – System)



Slot Description		Slot Description		
	Default	Option 1*	Option 2**	
1	PCIe 5.0 x16 FHFL	PCIe 5.0 x8 FHFL	PCIe 5.0 x16 FHFL	1
2	---	PCIe 5.0 x8 FHFL	---	2
3	PCIe 5.0 x16 FHFL	PCIe 5.0 x8 FHFL	PCIe 5.0 x16 FHFL	3
4	---	PCIe 5.0 x8 FHFL	---	4
Slot Description		Slot Description		
5	AIOM / OCP NIC 3.0 Slot	AIOM / OCP NIC 3.0 Slot		
6	AIOM / OCP NIC 3.0 Slot	AIOM / OCP NIC 3.0 Slot		

CPU1 CPU2

*For slot 1 - 4 Option 1 config, two optional cables are required to replace existing cables.
**For slot 1 - 4 Option 2 config, switching from (a) RSC to (b) RSC is required.

System Cooling	Fans: Up to 6 Counter-rotating, Hot-swappable 6cm Fan(s) Air Shroud: 4 Air Shrouds
Power Supply	2x 1300W Redundant power supplies
System BIOS	BIOS Type: AMI 256MB SPI Flash
Management	SuperCloud Composer®; Supermicro Server Manager (SSM); Supermicro Update Manager (SUM); Supermicro SuperDoctor® 5 (SD5); Super Diagnostics Offline (SDO); Supermicro Thin-Agent Service (TAS); SuperServer Automation Assistant (SAA) New!
PC Health Monitoring	CPU: Monitors for CPU Cores, Chipset Voltages, Memory FAN: Fans with tachometer monitoring Pulse Width Modulated (PWM) fan connectors Status monitor for speed control Temperature: Monitoring for CPU and chassis environment Thermal Control for fan connectors
Dimensions and Weight	Weight: Gross Weight: 47.4 lbs (21.5 kg) Net Weight: 34.4 lbs (15.6 kg) Available Color: Silver
Operating Environment	Operating Temperature: 0°C to 35°C (32°F to 95°F) Non-operating Temperature: -40°C to 70°C (-40°F to 158°F) Operating Relative Humidity: 8% to 90% (non-condensing) Non-operating Relative Humidity: 5% to 95% (non-condensing)
Motherboard	Super X13DEM
Chassis	CSE-HE211-R000NFP